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Bioconductor

consICA: a package for reference-free deconvolution reveals oncogenic processes in "omics" data and highlights the benefits of multimodal data analysis for patient stratification

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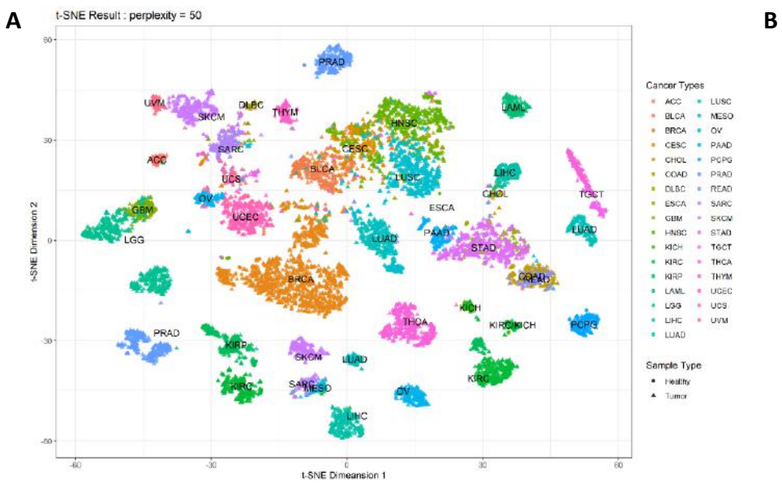
Challenges

In cancer research, the analysis of patient-derived molecular profiles, such as gene expression or DNA methylation, is not straightforward because of the tumor heterogeneity that masks important molecular signatures. This can be caused by natural variability in cell type proportions or by clonal variation between species of tumor cells. Additionally, technical biases between experimental platforms may limit the direct comparison of patient data to large public datasets.

Data

TCGA
> 11k patients
33 types of tumors
clinical data
(age, gender, survival...)
mRNA
(10k samples, 20k features)
miRNA
(> 9k samples, ~1k features)
methylation
(>9k samples, 450k features)



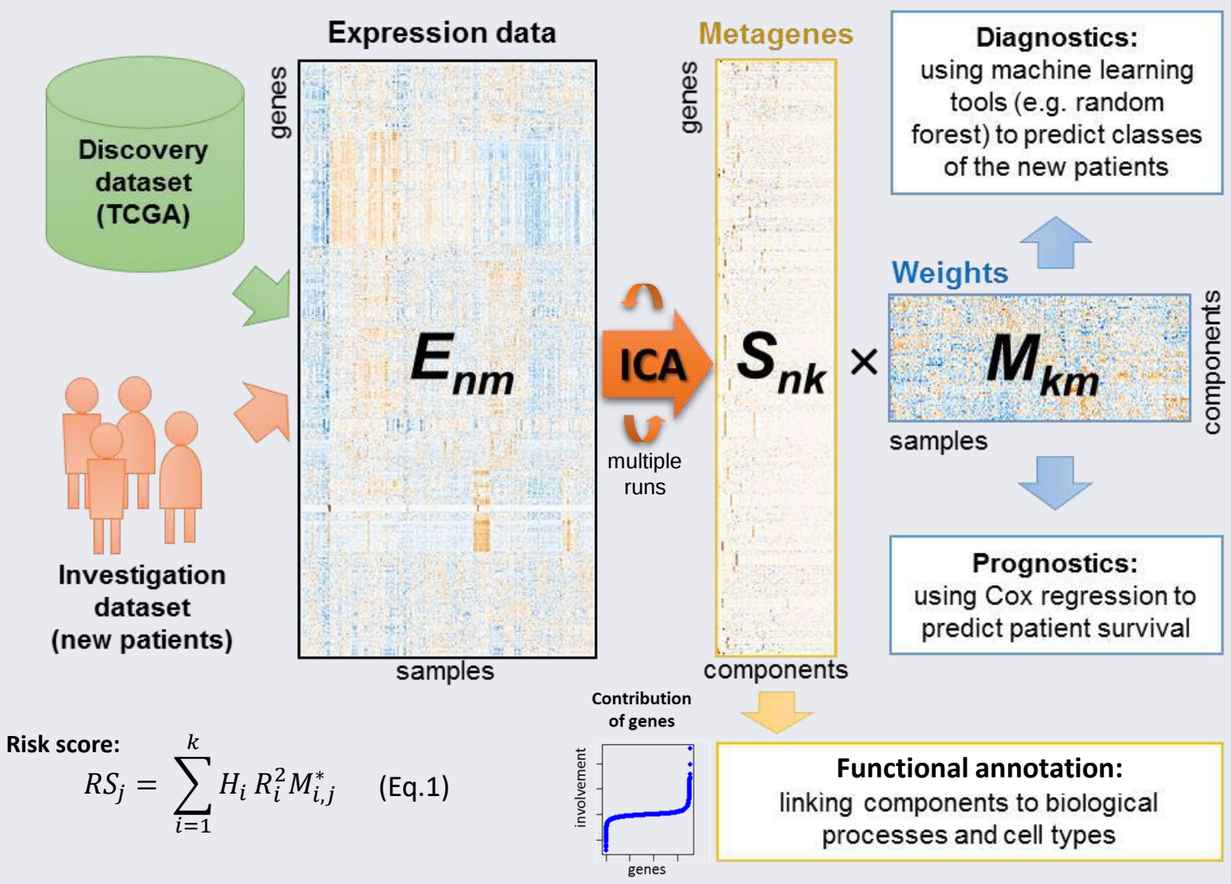
B

LGG	Brain Lower Grade Glioma
KIRP	Kidney renal papillary cell carcinoma
KIRC	Kidney renal clear cell carcinoma
ACC	Adrenocortical carcinoma
SARC	Sarcoma
UVM	Uveal Melanoma
MESO	Mesothelioma
SKCM	Skin Cutaneous Melanoma
LIHC	Liver hepatocellular carcinoma
PAAD	Pancreatic adenocarcinoma
LAML	Acute Myeloid Leukemia
LUAD	Lung adenocarcinoma
BRCA	Breast invasive carcinoma
HNSC	Head and Neck squamous cell carcinoma
BLCA	Bladder Urothelial Carcinoma
CESC	Cervical squamous cell carcinoma and endocervical adenocarcinoma
UCEC	Uterine Corpus Endometrial Carcinoma
KICH	Kidney Chromophobe
GBM	Glioblastoma multiforme
LUSC	Lung squamous cell carcinoma
THCA	Thyroid carcinoma
CHOL	Cholangiocarcinoma
UCS	Uterine Carcinosarcoma
DLBC	Lymphoid Neoplasm Diffuse Large B-cell Lymphoma
ESCA	Esophageal carcinoma
PCPG	Pheochromocytoma and Paraganglioma
PRAD	Prostate adenocarcinoma
STAD	Stomach adenocarcinoma
READ	Rectum adenocarcinoma
THYM	Thymoma
COAD	Colon adenocarcinoma
OV	Ovarian serous cystadenocarcinoma
TGCT	Testicular Germ Cell Tumors

FIGURE 2. TCGA datasets t-SNE visualization (A). Codes of the considered cancer datasets (B).

Method

https://bioconductor.org/packages/consICA

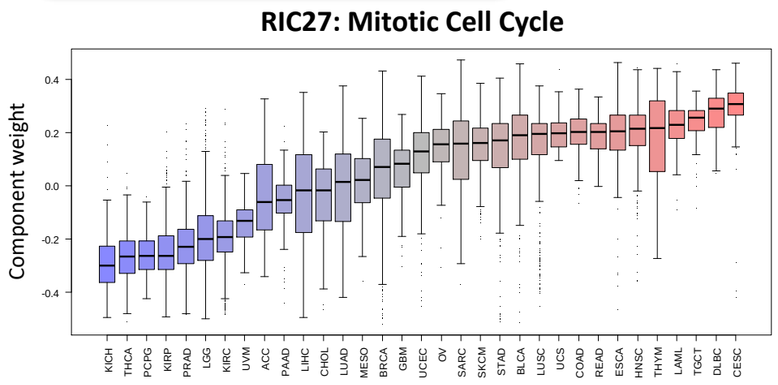


Risk score:

$$RS_j = \sum_{i=1}^{k} H_i R_i^2 M_{i,j}^* \quad \text{(Eq.1)}$$

FIGURE 1. Independent component analysis (ICA) [2-4] decomposes combined gene expression matrix from multiple bulk samples into meaningful signals in the space of genes (**S**) and weights in the space of samples (**M**). Biological processes and signatures of cell subtypes can be found in **S**, while **M** could be linked to patient groups and patient survival (Cox regression and Eq.1).

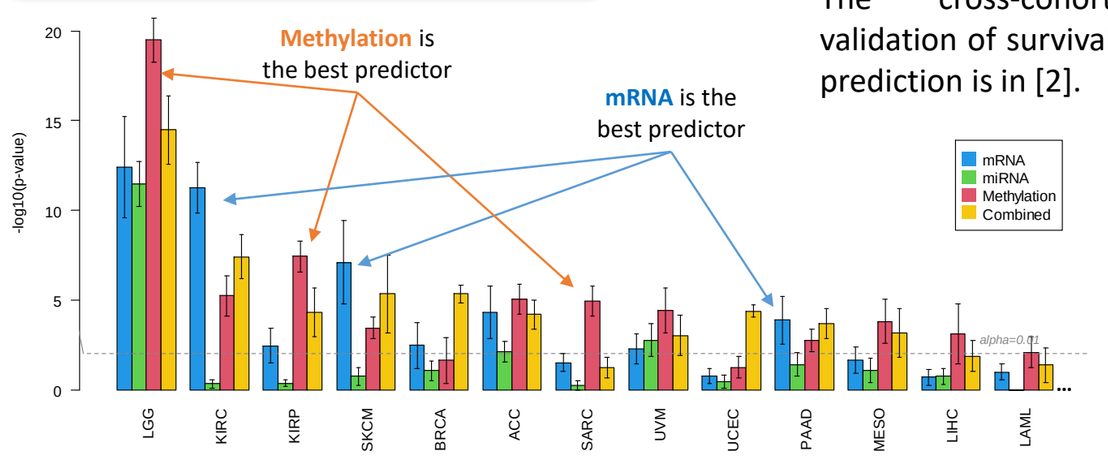
Biological processes



See [2] for the application of consICA in melanoma. Its application to gliomas is in [6]. Here we show a pan-cancer cell-cycle characterization.

FIGURE 2. Weights of components corresponding to mitotic cell cycle biological process.

Diagnostics & prognostics



The cross-cohort validation of survival prediction is in [2].

FIGURE 3. Prediction of survival based on IC risk score (Eq. 1) for TCGA data (same cohort, cross-validation).

Multi-omics data integration

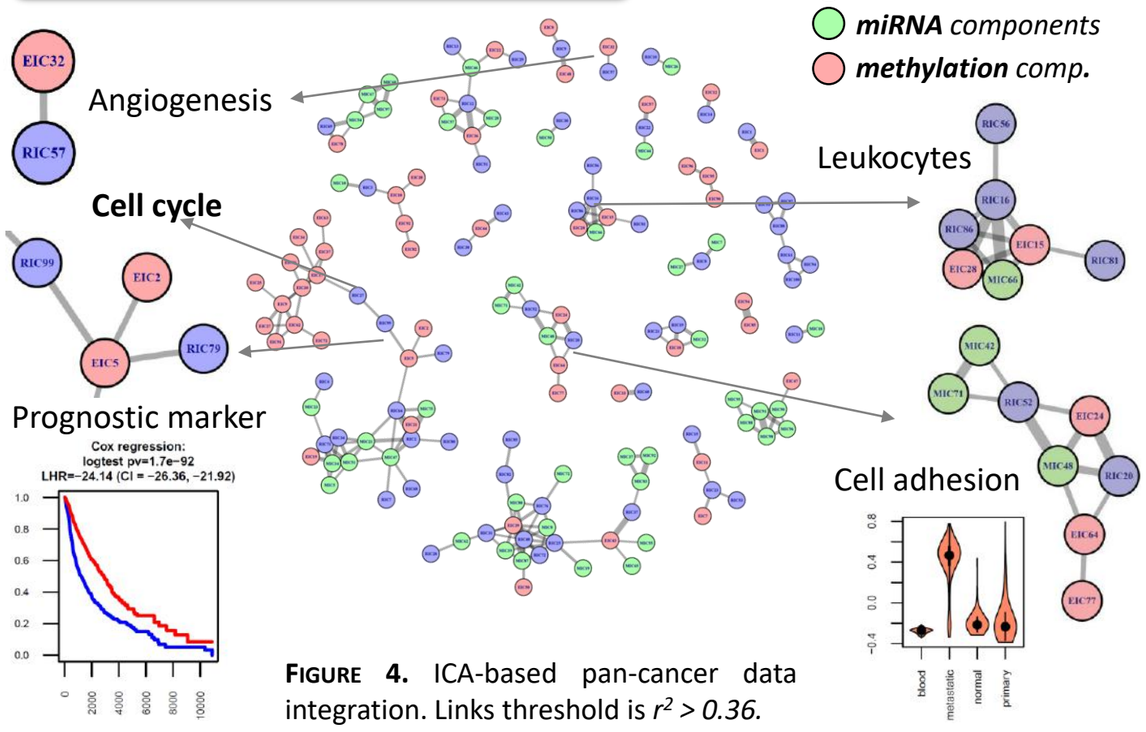


FIGURE 4. ICA-based pan-cancer data integration. Links threshold is $r^2 > 0.36$.

Conclusions

- We propose **consICA** – a Bioconductor (v.3.16) package [1].
- consICA **corrects technical biases** between united datasets [2,5,6].
- The method estimates the activity of **biological processes** in the new samples and their **cell composition**. [2-4,6]. It also can be used for **data integration** when several modalities of omics data are available [2,7].
- ICA generates features (weights) that can be used to **predict patient survival** or **diagnostic classes** [2,6]. Patients with different cancer types are better characterized by specific omics data (Fig.3).

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References

1. **https://bioconductor.org/packages/consICA**
2. Nazarov et al., **2019, BMC Med Gen** 12
3. Sompairac et al., **2019, Int J Mol Sci** 20(18)
4. Captier et al., **2022**, **Bioinformatics** 38(10)
5. Scherer et al., **2020**, **Nat Protoc** 15
6. Golebiewska et al., **2020, Acta Neuropathol** 140(16)
7. Nazarov, Kreis, **2021, Comp Struct Biotechnol J**, 19

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